PROJECT DOCUMENTATION

Student Performance Analytics System

Python with AI Mini Project

Project Title	Student Performance Analytics System
Technology	Python, Pandas, NumPy, Matplotlib
Platform	Google Colab
Dataset	Custom dataset with 30 student records
GitHub	github.com/Deepaliiiii/student-performance-analytics

This document describes the objective, dataset, implementation, results, visualizations, and learning outcomes.

1. Objective

Build a Python-based system that reads student data and analyzes it using Pandas and NumPy to calculate marks summaries, averages, grades, pass/fail statistics, subject-wise performance, and top performers.

2. Technologies Used

Technology	Purpose
Python	Core programming
Pandas	Load and analyze tabular data
NumPy	Numerical calculations
Matplotlib	Graphs and pictorial representations
Google Colab	Development environment
GitHub	Project repository

3. Dataset Description

The custom students.csv dataset contains 30 records. Fields include Student_ID, Name, Department, Python, Mathematics, DBMS, AI, and Attendance. Total_Marks, Average_Marks, Grade, and Result are calculated in the notebook.

4. Implementation

• **Data loading:** Read the CSV into a Pandas DataFrame and inspect columns, record count, data types, and descriptive statistics.

• **Marks calculations:** Use NumPy to calculate total marks across four subjects and calculate each student's average.

• **Grades:** Use a reusable assign_grade() function.

• **Pass/fail:** Use check_result() with average and attendance conditions.

• **Analysis:** Find highest/lowest average, class average, subject averages, pass/fail counts, and top five students.

• **Visualization:** Use Matplotlib to create four graphs.

5. Grading and Pass/Fail Rules

Average Marks	Grade
90–100	A+
80–89.99	A
70–79.99	B
60–69.99	C
50–59.99	D
Below 50	F

Pass condition: Average marks must be at least 40 AND attendance must be at least 75%. Otherwise, the student is classified as Fail.

6. Key Features

• Pandas DataFrame data handling

• NumPy numerical calculations

• Reusable functions, loops, and conditions

• Total and average marks for each student

• Grade and pass/fail assignment

- Highest, lowest, and class-average analysis
- Subject-wise averages and best subject
- Top five student leaderboard
- Pass/fail statistics
- Four data visualizations

7. Analysis Results

Analysis	Result
Number of students	30
Highest performer	Tanvi (ST028), average 95.25
Lowest performer	Sahil (ST017), average 49.75
Overall class average	75.81
Students passed	24 (80%)
Students failed	6 (20%)
Best-performing subject	DBMS, average 77.23
Top five students	Tanvi, Meera, Priya, Shreya, Aditya

8. Practical Questions Answered

Which student has the highest average? Tanvi (ST028), with 95.25.

Which subject has the best average? DBMS, with 77.23.

How many students passed? 24 passed and 6 failed.

What is the overall class average? 75.81 marks.

9. Visualizations

Screenshots below are from the completed Google Colab notebook.

Figure 1. Average Marks by Subject

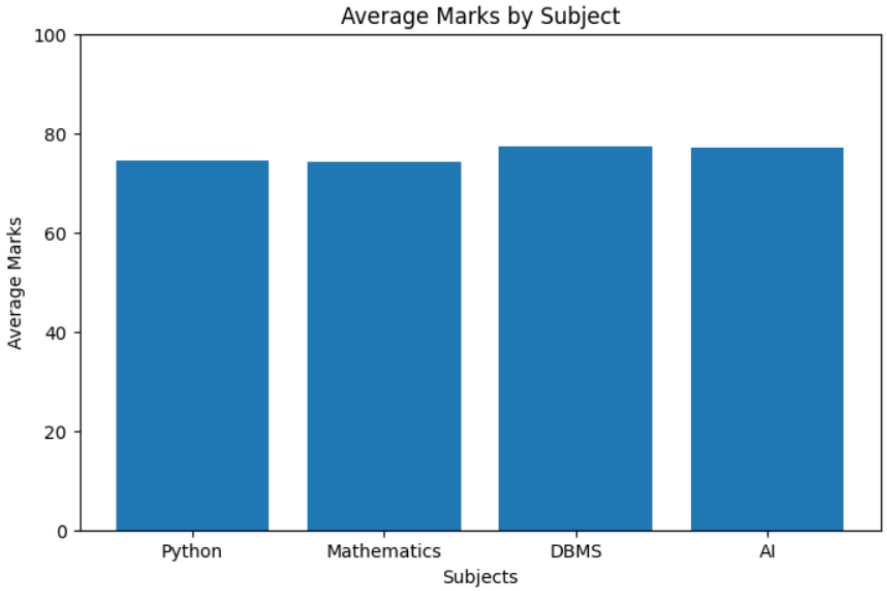


Figure 2. Top 5 Students by Average Marks

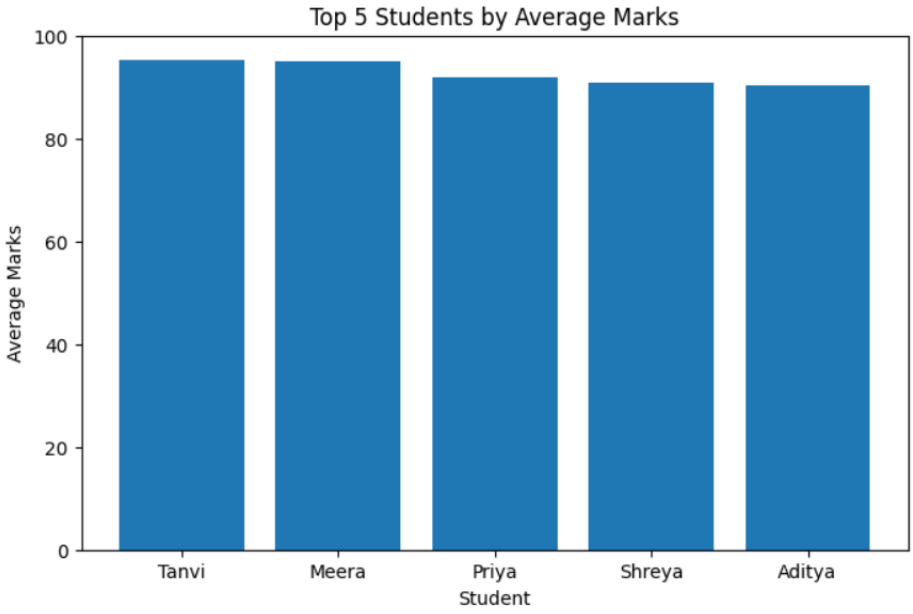

Top 5 Students by Average Marks
Average Marks
100
80
60
40
20
0
Tanvi
Meera
Priya
Shreya
Aditya
Student

Figure 3. Student Pass/Fail Distribution

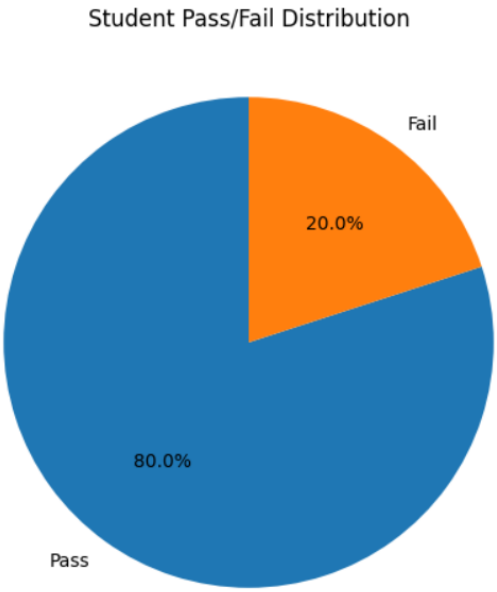
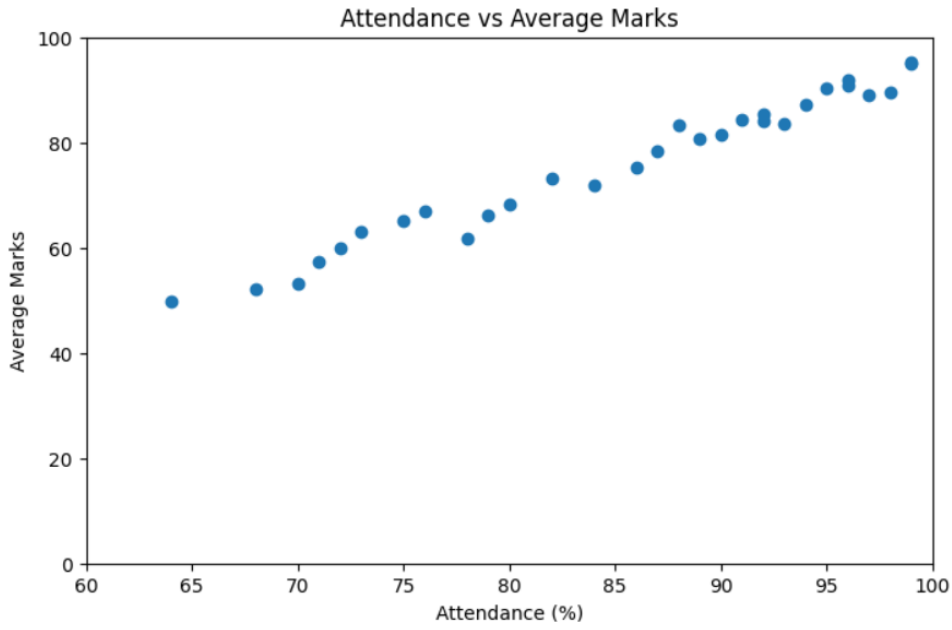


Figure 4. Attendance vs Average Marks



10. Final Outcome

The project provides a complete data-analysis workflow: loading student data, calculating academic metrics, assigning grades and results, identifying performance patterns, answering practical questions, and presenting findings through graphs.

11. Challenges and Learning

The project provides practical experience with Pandas DataFrames, NumPy calculations, reusable functions, loops, conditions, sorting and summarizing data, and Matplotlib visualizations. It demonstrates how raw CSV records can be converted into understandable information.

12. GitHub Repository

https://github.com/Deepaliiiii/student-performance-analytics

The repository should contain the Colab notebook, students.csv, README.md, and this documentation PDF.

13. Final Submission Checklist

■ Notebook uploaded and runs successfully

■ Dataset included

■ NumPy and Pandas used

■ Functions, loops, and conditions included

■ Four visualizations included

■ README.md included

■ Documentation PDF uploaded

■ Repository checked

■ GitHub URL submitted through the course Google Form